

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Volume & Surface Area

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

VOLUME & SURFACE AREA · CH4 · L7

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The Map

This lesson collapses Volume & Surface Area into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Name the solid

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Use prism volume

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Unfold surface area

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Reverse from volume or area

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Use trigonometry

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 39 website questions, 39 checked solutions, 4 local PDFs read.

01 Name the solid

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Volume & Surface Area question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{Area of cross-section} = \frac{1}{2}(37 + 28)(20) = 650$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P2H Q17

The cross section is a

$$\text{Area of cross-section} = \frac{1}{2}(37 + 28)(20) = 650$$

Calculate area

$$\frac{37 - 28}{2} = 4.5$$

Finish: 3844 cm², and 39.3°.

FORMULA BANK

Prism volume = area of cross-section × length. Surface area = sum of visible faces.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

17 The diagram shows a solid prism ABCDEFGH. A 28 cm 24 cm 20 cm D E F G H B C Diagram NOT accurately drawn 37 cm The trapezium ABCD, in which AD is parallel to BC, is a cross section of the prism. The base ADEH of the prism is a horizontal plane. ADEH and BCFG are rectangles. The midpoint of BC is vertically above the midpoint of AD so that BA = CD. AD = 37 cm GF = 28 cm DE = 24 cm The perpendicular distance between edges AD and BC is 20 cm. (a) Work out the total surface area of the prism. cm² (4) (b) Calculate ...

YOUR SOLUTION

02 Use prism volume

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Volume & Surface Area question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{1}{2}(7 + 12)h = 9.5h$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
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WORKED MODEL

Source pattern: Jan 2020 P2HR Q13

Area of the trapezium cross

$$\frac{1}{2}(7 + 12)h = 9.5h$$

Volume of the prism

$$9.5h \times 10 = 608$$

Finish: $h = 6.4$ cm.

FORMULA BANK

Prism volume = area of cross-section $\times$ length. Surface area = sum of visible faces.

— BOTTOM LINE

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Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

13 Diagram NOT accurately drawn A B D C 7 cm E 10 cm 12 cm h cm F G H The diagram shows a prism ABCDEFGH in which ABCD is a trapezium with BC parallel to AD and CDEF is a rectangle. BC = 7 cm AD = 12 cm DE = 10 cm The height of trapezium ABCD is h cm The volume of the prism is 608 cm³ Work out the value of h. h =

YOUR SOLUTION

03 Unfold surface area

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Volume & Surface Area question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{1}{3}\pi r^2 h = 1600$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q26

Let the radius of the

This is a reasoning step: write one clear sentence, then continue the calculation.

The volume of the cone

$$\frac{1}{3}\pi r^2 h = 1600$$

Finish: 107°

FORMULA BANK

Prism volume = area of cross-section $\times$ length. Surface area = sum of visible faces.

— BOTTOM LINE

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Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

26 Here is a sector, AOB, of a circle with centre O and angle AOB = x degrees. Diagram NOT accurately drawn. The sector can form the curved surface of a cone by joining OA to OB. Diagram NOT accurately drawn. The height of the cone is 25 cm. The volume of the cone is 1600 cm³. Work out the value of x . Give your answer correct to the nearest whole number. ' x =

YOUR SOLUTION

04 Reverse from volume or area

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Volume & Surface Area question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{1}{3}\pi r^2 h = 1600$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
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- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q26

Let the radius of the

This is a reasoning step: write one clear sentence, then continue the calculation.

The volume of the cone

$$\frac{1}{3}\pi r^2 h = 1600$$

Finish: 107°

FORMULA BANK

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BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

26 Here is a sector, AOB, of a circle with centre O and angle AOB = x degrees. Diagram NOT accurately drawn. The sector can form the curved surface of a cone by joining OA to OB. Diagram NOT accurately drawn. The height of the cone is 25 cm. The volume of the cone is 1600 cm³. Work out the value of x . Give your answer correct to the nearest whole number. ' x =

YOUR SOLUTION

05 Use trigonometry

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Volume & Surface Area question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{Area of cross-section} = \frac{1}{2}(37 + 28)(20) = 650$$

SIMPLEST STRATEGY

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WORKED MODEL

Source pattern: Jan 2021 P2H Q17

The cross section is a

$$\text{Area of cross-section} = \frac{1}{2}(37 + 28)(20) = 650$$

Calculate area

$$\frac{37 - 28}{2} = 4.5$$

Finish: 3844 cm², and 39.3°.

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Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

17 The diagram shows a solid prism ABCDEFGH. A 28 cm 24 cm 20 cm D E F G H B C Diagram NOT accurately drawn 37 cm The trapezium ABCD, in which AD is parallel to BC, is a cross section of the prism. The base ADEH of the prism is a horizontal plane. ADEH and BCFG are rectangles. The midpoint of BC is vertically above the midpoint of AD so that BA = CD. AD = 37 cm GF = 28 cm DE = 24 cm The perpendicular distance between edges AD and BC is 20 cm. (a) Work out the total surface area of the prism. cm² (4) (b) Calculate ...

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Name the solid
"work out", "show", "hence", a familiar exam phrase	Use prism volume
"work out", "show", "hence", a familiar exam phrase	Unfold surface area
"work out", "show", "hence", a familiar exam phrase	Reverse from volume or area
"work out", "show", "hence", a familiar exam phrase	Use trigonometry

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the

trigger, write the first line, and let the working follow.